

CSE 470 – Class Diagram Design

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Guidelines for Analyzing Requirements / Use Cases

- A common or improper noun implies a class of objects
- A proper noun implies an instance of a class
- A collective noun implies a class of objects made up of groups of instances of another class



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Guidelines for Analyzing Requirements / Use Cases (2)

- An adjective implies an attribute of an object
- A doing verb implies an operation
- A being verb implies a relationship between an object and its class
- A having verb implies an aggregation or association relationship



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Guidelines for Analyzing Requirements / Use Cases (3)

- A transitive verb implies an operation
- An intransitive verb implies an exception
- A predicate or descriptive verb phrase implies an operation
- An adverb implies an attribute of a relationship or an operation



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Class Diagram

- Ensure that the classes are both necessary and sufficient to solve the underlying problem
  - no missing attributes or methods in each class
  - no extra or unused attributes or methods in each class
  - no missing or extra classes



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Discarding Unnecessary and Incorrect Classes

- Redundant Classes: Some potential classes differ only in name.
- Irrelevant Classes: Classes that have nothing to do with the system. Example: *computer connection*
- Vague Classes: Classes whose meaning is not clear at all. Examples: *system* and *software*



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Discarding Unnecessary and Incorrect Classes

- Attributes: Some nouns in the list above are likely to be modeled as attributes rather as classes. Examples: *author*, *title*
- Operations: Some nouns are likely to be operations rather than classes. Example: *book search*.
- Roles: Some nouns are roles of objects involved in associations rather than classes.
- Implementation Constructs: Anything that is not part of the real-world problem.



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Types of Classes

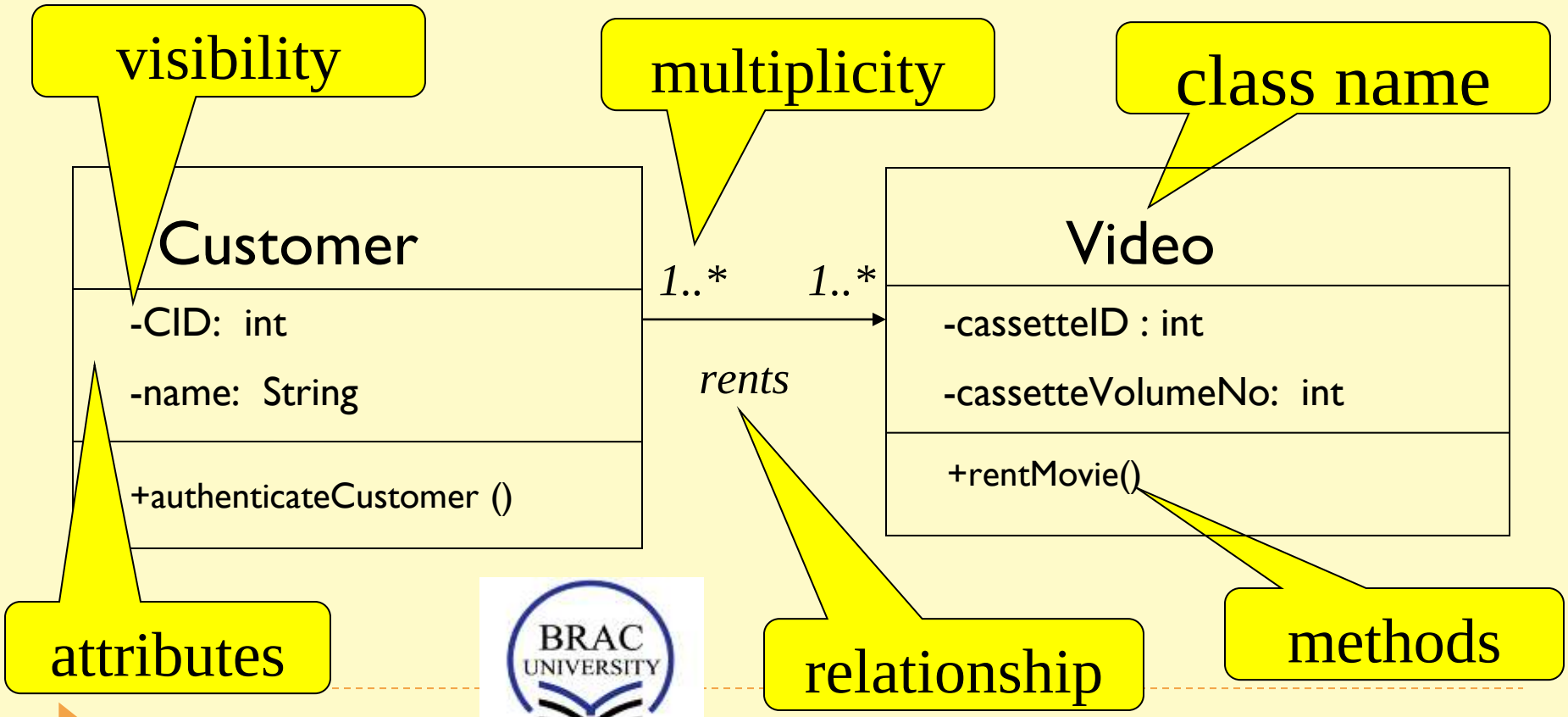
- Ones found during analysis:
  - people, places, events, and things about which the system will capture information
  - ones found in application domain
- Ones found during design
  - specific objects like windows and forms that are used to build the system



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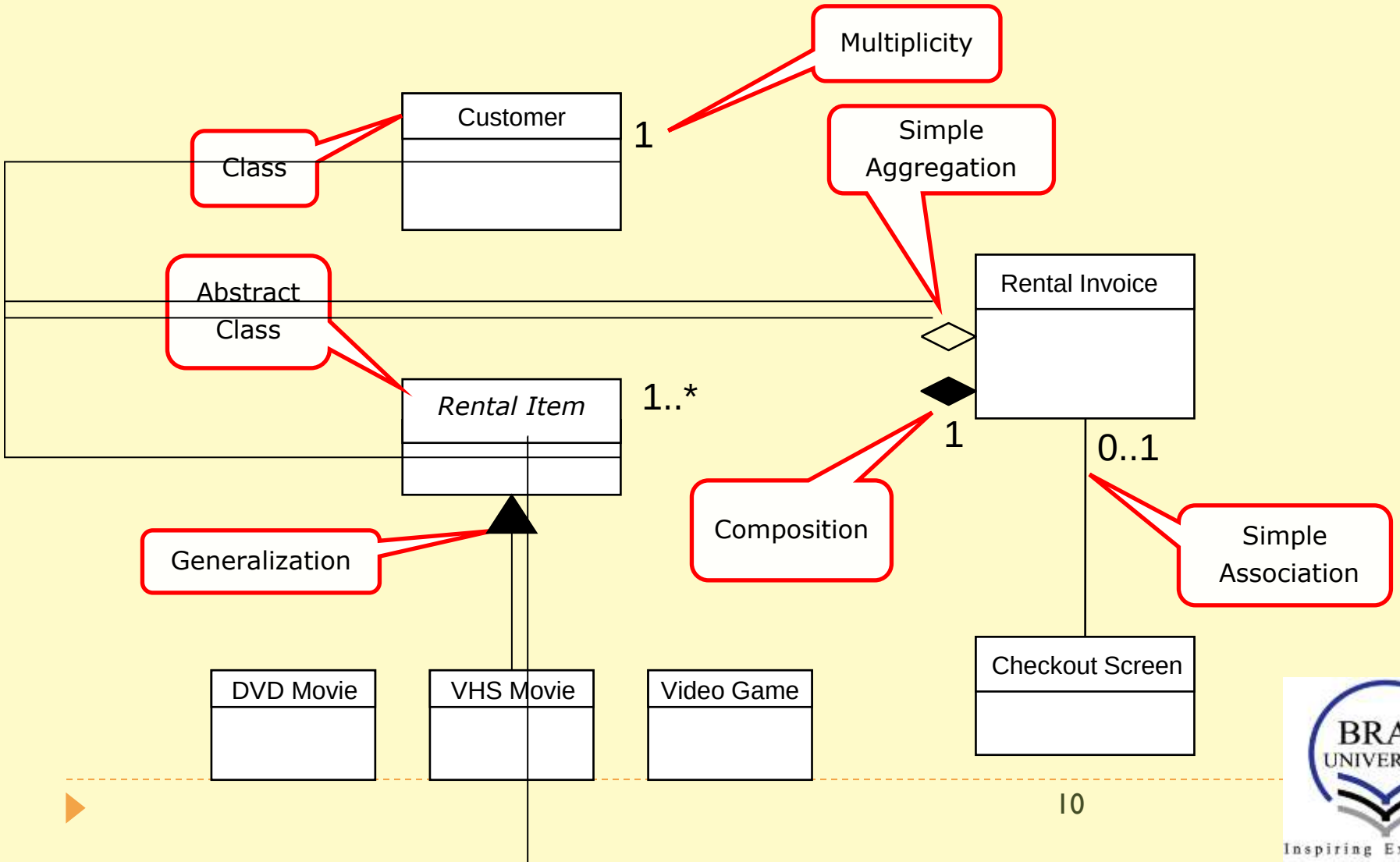
Example of a Class Diagram

Video Rental System



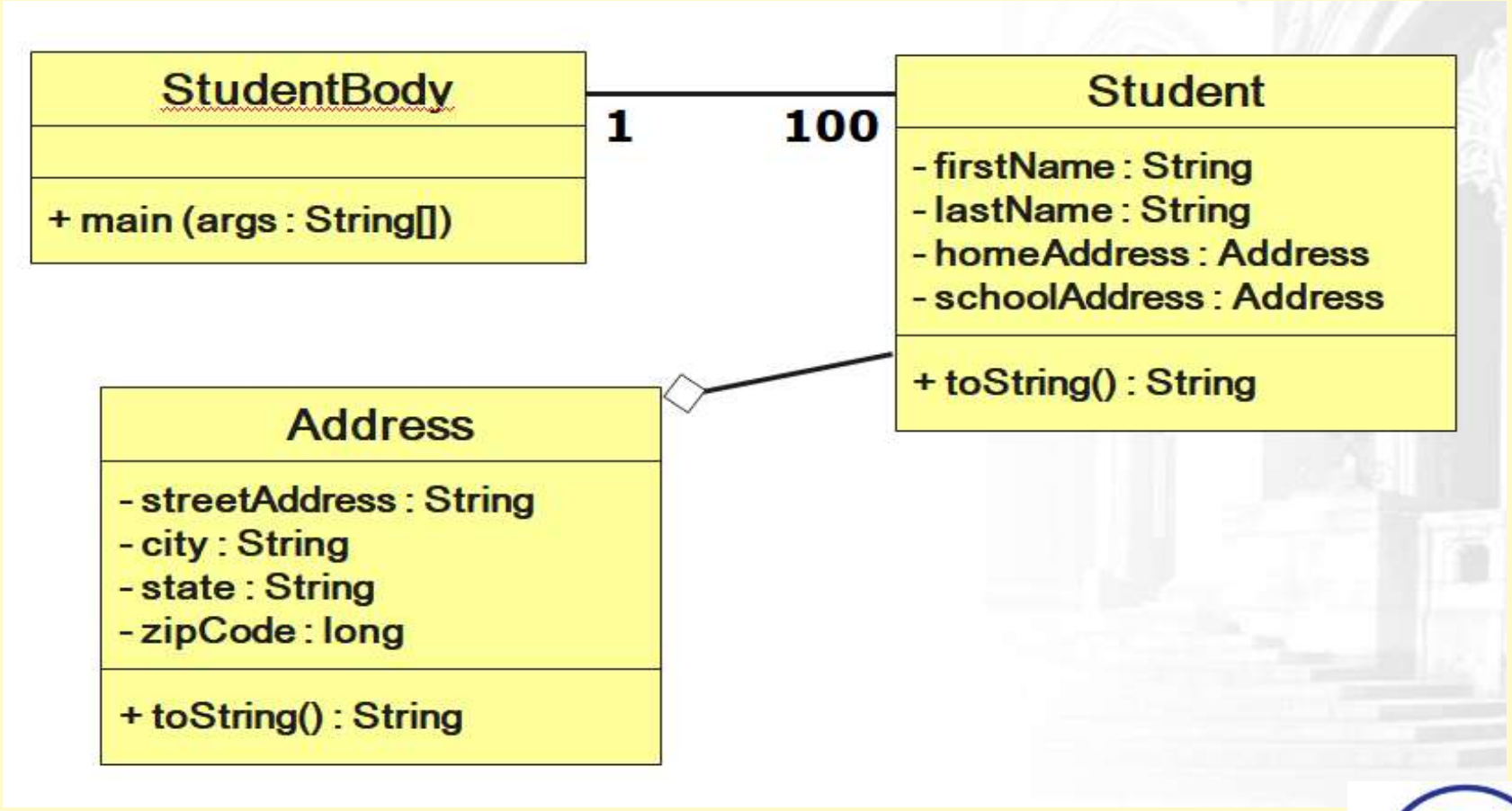
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Class diagram example 1

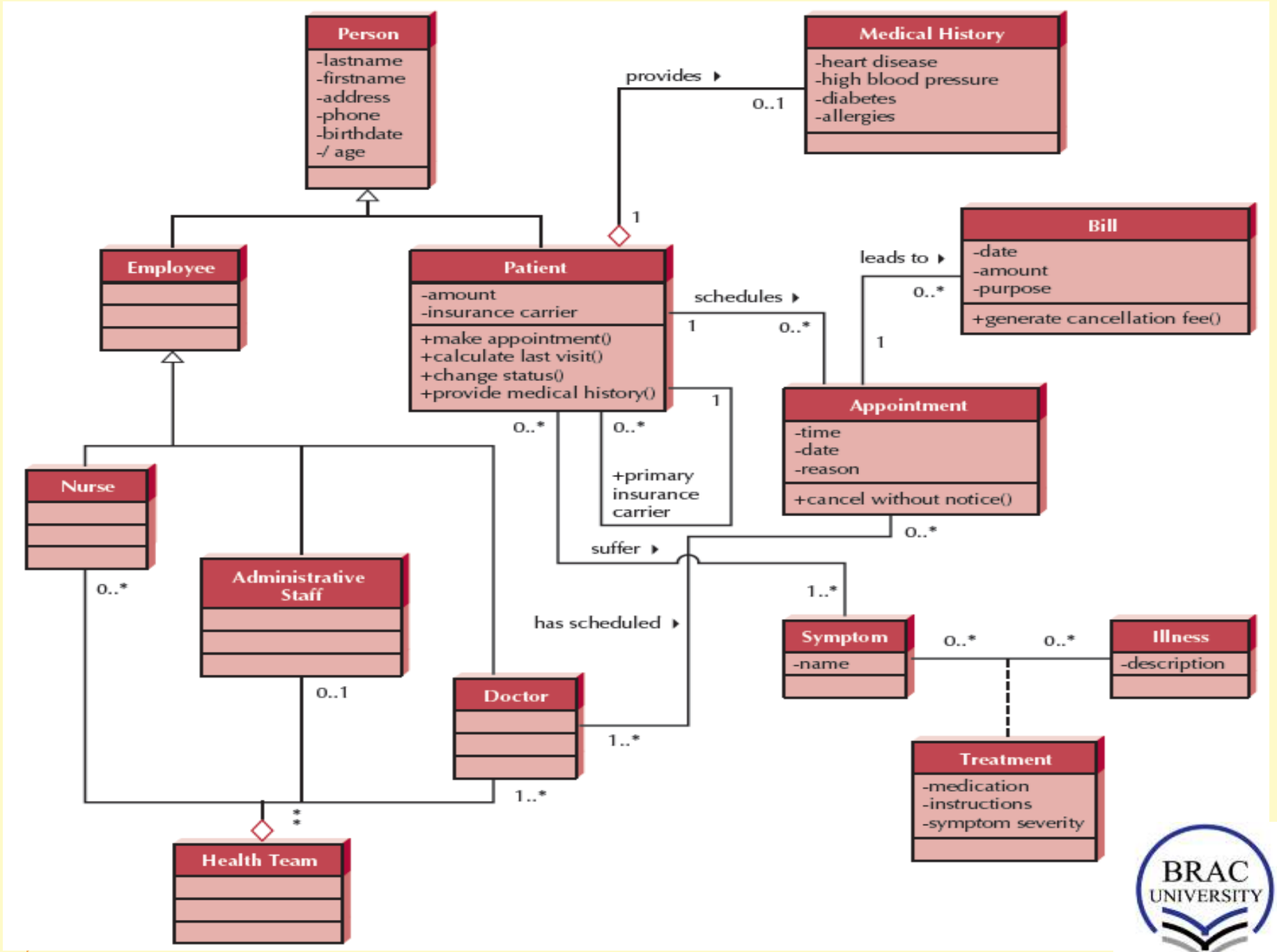


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Class diagram example 2



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Person
-lastname
-firstname
-address
-phone
-birthdate
-/ age
Medical History
-heart disease
-high blood pressure
-diabetes
-allergies
provides
0..1
1
Employee
Patient
-amount
-insurance carrier
+make appointment()
+calculate last visit()
+change status()
+provide medical history()
schedules
1
0..*
Bill
-date
-amount
-purpose
+generate cancellation fee()
leads to
0..*
1
Appointment
-time
-date
-reason
+cancel without notice()
0..*
0..*
1
+primary insurance carrier
Nurse
0..*
Administrative Staff
0..1
suffer
1..*
has scheduled
Symptom
-name
0..*
0..*
Illness
-description
Doctor
1..*
1..*
Treatment
-medication
-instructions
-symptom severity
Health Team
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